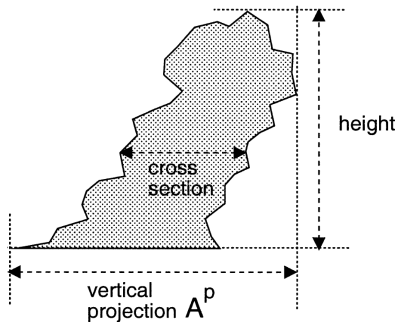


Size Statistics of Cumulus Cloud Populations in Large-Eddy Simulations



Three shallow-cumulus cases

BOMEX over the ocean.

SCMS and ARM over land.

A cloud is a connected set of saturated grid cells.
Its size is

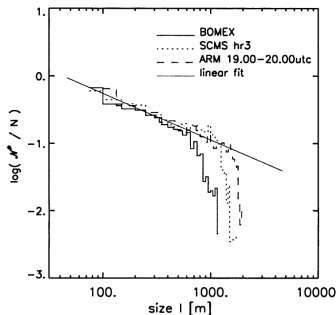
$$l = \sqrt{A^p}$$

A^p : vertically projected area.

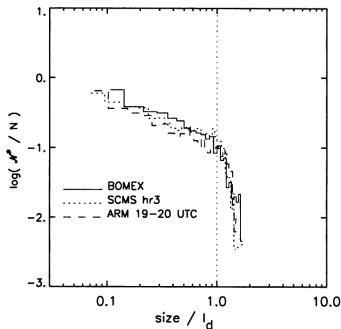
The study measures cloud counts, cloud cover and mass flux by cloud size

A power law with a case-dependent size break

2



Cloud size l



Rescaled size l/l_d

Below the break

$$\mathcal{N}(l) \propto l^{-1.70}$$

About one decade of sizes.

Break scale ℓ_d

BOMEX: 700 m

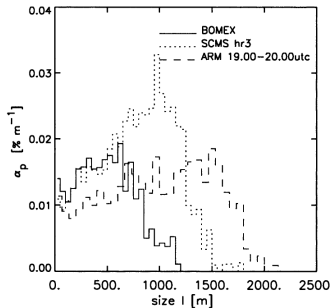
SCMS: 1050 m

ARM: 400–1250 m

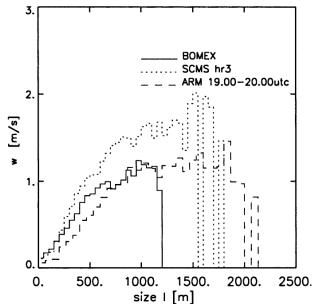
Rescaling by ℓ_d approximately collapses the three cloud-size densities

Intermediate cloud sizes dominate vertical transport

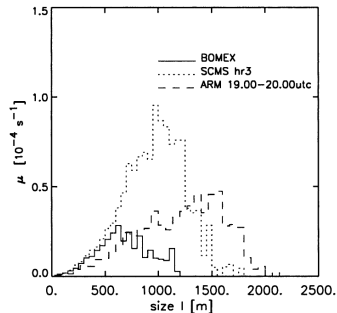
3



Cover peaks near the size break.



Vertical velocity grows with cloud size.



Mass flux peaks at somewhat larger sizes.

The smallest clouds contribute little to the total vertical mass transport

What is missing for a 2D thermal field

4

No distributions reported for

Thermal size at a specified height.

Thermal positions in the (x, y) plane.

Distances between thermals.

The cloud mask does not identify dry thermals below cloud base.

Possible use: our inference

The cloud-size break ℓ_d suggests a scale to compare with cloud-feeding updraft sizes.

Test this with paired cloud and updraft measurements at a fixed height.

Neither -1.70 nor ℓ_d transfers directly to a thermal-size law.

Cloud-size statistics offer a hypothesis to test, with no thermal spacing law